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**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE,
PILANI
II SEMESTER 2025-26**

CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing

24-04-2026 (Friday)

QUIZ-6 (closed book)

Max Marks: 08

Max time: 15 Min

B

Q1. An 8086-based interrupt-driven control system uses three 8259A PICs in cascaded mode: one Master PIC and two Slave PICs. The Master PIC is initialized with vector base address **40H**. Slave-1 is connected to IR2 of the Master and initialized with vector base address **70H**. Slave-2 is connected to IR5 of the Master and initialized with vector base address **A0H**. During operation, the following interrupt requests occur simultaneously:

- IR4 of Master
- IR6 of Slave-1
- IR3 of Slave-2
- Assume fully nested mode, edge-triggered interrupts, and no masking. Answer the following:

Question	Answer	
1. a. What is the vector number of IR6 of Slave-1 (mention in hex) [2M]	76h	
1.b Which interrupt request will be serviced first? .[2M]	IR6 of Slave 1	
3. If the ISR address for IR3 of Slave-2 is stored as IP = 2500H and CS = 9000H, write the four IVT memory locations and their contents for IR3 of slave 2. [2M]	Address (in hex)	Content (in hex)
	0028C	00
	0028D	25
	0028E	00
	0028F	90

Q.2. An 8086-based system uses an 8254 timer. Counter 0 is programmed in Mode 2 (Rate Generator) with a clock input of 1 MHz. The counter is loaded with a count value of $N = 1000$. The output of Counter 0 is connected to a device that is sensitive to the duty cycle of the output waveform. Ignore the slight variation in duty cycle if any in the first period.

Question	Answer
1.a What is the duration of LOW time of the output signal? (mention in microseconds) [1M]	1
1.b What is the duration of HIGH time of the output signal? (mention in microseconds) [1M]	999